

Superstore SQL Analysis — Project Report

1. Executive Summary

This project analyzes Superstore sales data using SQL Server. The workflow starts with raw Excel-based source tables and transforms them into a relational structure containing customers, orders, order details, products, and locations.

The analysis combines database design, data validation, multi-table joins, advanced SQL techniques, KPI reporting, reusable views, and stored procedures.

The final dataset contains **2,323 order-detail rows**, **1,175 distinct orders**, and **629 customers**. Total sales were **501,239.75**, with total profit of **39,706.21** and an overall profit margin of **7.92%**.

The results show a clear difference between sales volume and profitability: Consumer customers generated the most sales, while Corporate customers generated the most profit. Technology was the strongest product category, while Furniture produced a negative total profit despite generating substantial sales.

2. Project Objectives

The project objectives were to:

1. Build a relational SQL Server database from raw Excel tables.
 2. Create primary and foreign key relationships.
 3. Handle source-data quality and key-uniqueness issues.
 4. Validate the transformed data.
 5. Analyze customer, shipping, product, sales, profit, and regional performance.
 6. Use advanced SQL concepts such as subqueries and CTEs.
 7. Create reusable views and stored procedures.
 8. Produce KPI-level business insights.
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3. Data Preparation & Modeling

Customers

The customer source was transformed into a `Customers` table containing:

- Customer ID
- Customer Name
- Segment

`Customer ID` was defined as the primary key.

Products

The source `Product ID` was found to be non-unique. Because a primary key must uniquely identify each row, `Product ID` was not used as the final primary key.

A surrogate `ProductKey` was created using:

```
ALTER TABLE Products  
ADD ProductKey INT IDENTITY(1,1);
```

`ProductKey` became the primary key and was subsequently used to link `Orders_Details` to `Products`.

Orders

The `Orders` table was created using distinct order-level records.

Rows with a null `Order ID` were removed before applying the primary-key constraint.

Location

A `LocationKey` surrogate key was added to the location table.

The source location data did not contain a direct business key linking it to individual order-detail rows. A temporary row-number mapping mechanism was therefore used during transformation, followed by a validation query.

Orders_Details

Orders_Details contains the transaction-level measures:

- Sales
- Quantity
- Discount
- Profit

It is linked to:

- Orders through Order ID
 - Products through ProductKey
 - Location through LocationKey
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4. Data Validation

The final validation returned:

```
TotalOrderDetails = 2323
RowsWithProduct  = 2323
RowsWithLocation = 2323
```

Therefore, all order-detail rows had valid product and location mappings at the end of the transformation.

5. KPI Results

KPI Result

Total Sales 501,239.75
Total Profit 39,706.21
Total Orders 1,175
Total Customers 629
Total Quantity 8,780
Profit Margin 7.92%

Interpretation

The project generated approximately **501.2K in sales** and **39.7K in profit**, resulting in a **7.92% profit margin**.

6. Customer Analysis

Customers by Segment

Segment Customers

Consumer 328
Corporate 180
Home Office 121

Consumer customers represent the largest customer segment.

Sales & Profit by Segment

Segment Sales Profit

Consumer 252,031.38 8,563.97
Corporate 157,995.75 18,703.82
Home Office 91,212.62 12,438.42

Key Insight

Consumer generated the highest sales, but Corporate generated the highest profit.

This is an important business distinction: **the segment with the most revenue is not necessarily the segment creating the most profit.**

Top 10 Customers by Profit

The highest-profit customer was:

- **TC-20980 — 8,745.06**

Other high-performing customers included:

- AB-10105 — 5,362.61
- SC-20095 — 4,668.70

- AR-10540 — 2,602.09
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7. Shipping Analysis

Shipping Performance

Ship Mode Average Shipping Days Orders

Same Day 0 62

First Class 2 174

Second Class 3 224

Standard Class 4 715

Standard Class was by far the most frequently used shipping method, accounting for **715 orders**.

Yearly Average Shipping Time

Year Average Shipping Days

2013 4

2014 3

2015 4

2016 3

Average shipping time remained within a relatively narrow range of 3–4 days for yearly averages.

8. Sales & Profit Analysis

Performance by Category

Category Sales Profit

Technology 170,416.29 33,697.52

Office Supplies 167,026.32 8,879.78

Furniture 163,797.14 -2,871.09

Key Insights

Technology was the strongest category:

- Highest sales: 170,416.29
- Highest profit: 33,697.52

Furniture is a major area for investigation:

- Sales: 163,797.14
- Profit: -2,871.09

This means Furniture generated significant revenue but resulted in an overall loss.

9. Sub-Category Analysis

The project also analyzed sales and profit by sub-category.

Several sub-categories produced negative profit, including:

- Supplies: -661.89
- Binders: -1,043.69
- Machines: -1,486.07
- Bookcases: -1,997.93
- Appliances: -2,638.63
- Tables: -3,559.66
- Furnishings: -3,906.17

The results suggest that profitability should be evaluated at the sub-category level rather than relying only on total sales.

10. Discount Analysis

The analysis grouped sales and profit by discount level.

Notable results included:

Discount	Sales	Profit
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0.20	121,220.02	14,055.08
0.30	64,935.31	-6,540.89
0.40	19,546.22	-2,666.85
0.50	6,550.66	-4,309.74
0.60	6,644.68	-5,944.63
0.80	16,963.68	-30,539.09

Key Insight

Higher discount levels in this dataset are strongly associated with negative profit. The **0.80 discount level** is particularly concerning, with **-30,539.09** in profit.

This does not by itself prove that discounts caused the losses, but it identifies discount-heavy transactions as an important area for further investigation.

11. Monthly Performance

Across all years, monthly results were:

Month Sales Profit State

January	31,683.22	2,855.67	High Profit
February	8,211.28	-516.00	Loss
March	41,216.23	1,225.93	High Profit
April	26,177.32	11.95	Profitable
May	35,775.81	3,630.47	High Profit
June	33,081.24	2,184.78	High Profit
July	30,852.60	-2,217.96	Loss
August	26,846.19	2,685.93	High Profit
September	76,855.42	3,558.70	High Profit
October	53,186.63	12,644.29	High Profit
November	64,828.27	2,437.40	High Profit
December	72,525.54	11,205.05	High Profit

September had the highest monthly sales at **76,855.42**.

October had the highest monthly profit at **12,644.29**.

February and July were loss-making months in the combined monthly analysis.

12. Yearly Performance

Year Sales Profit State

2013 103,838.12 539.54 Profitable
2014 102,874.22 11,716.73 High Profit
2015 147,429.40 19,899.10 High Profit
2016 147,098.01 7,550.84 High Profit

Key Insight

2015 achieved the highest annual profit:

19,899.10

Sales were also highest in 2015 at **147,429.40**, slightly ahead of 2016.

13. Product Analysis

The project identified products with profit below the average product profit and also identified the highest-profit product within each category.

Highest-Profit Product by Category

Category Product Profit

Technology Canon imageCLASS 2200 8,399.98
Advanced Copier

Office Supplies GBC Ibimaster 500 3,804.90
Manual ProClick
Binding System

Furniture Hon Pagoda Stacking 770.35

Chairs

The Canon imageCLASS 2200 Advanced Copier was the strongest individual product by profit in its category.

14. Advanced SQL Analysis

CTE 1 — Top Customer by Segment

A two-step CTE calculates total customer sales and then finds the maximum sales value within each segment.

Results:

Segment Customer Sales

Consumer Adrian Barton 12,181.60

Corporate Tamara Chand 18,437.14

Home Office Tonja Turnell 5,301.68

CTE 2 — Monthly Sales vs Same-Year Average

The second CTE analysis compares each month's sales with the average monthly sales of its own year.

This allows the analysis to identify months that performed above or below their corresponding yearly monthly average.

15. Regional & City Analysis

The final regional query returned:

Region Sales Profit Orders

Central 501,239.75 39,706.21 1,175

The highest-performing cities were:

- **Houston** — highest sales: 80,629.77
- **Austin** — highest profit: 8,496.00

The difference between the highest-sales city and highest-profit city again demonstrates why revenue and profitability should be analyzed separately.

16. Reusable Database Objects

Views

vw_Monthly_Sales_Performance

Provides:

- Year
- Month
- Sales
- Profit
- Order count
- Profit margin

vw_Customer_Performance

Provides:

- Customer
- Segment
- Order count
- Total sales
- Total profit
- Average order value

Stored Procedures

sp_Sales_By_Date_Range

Accepts:

- Start date
- End date

and returns:

- Total sales
- Total profit
- Total orders
- Profit margin

sp_Regional_Performance

Accepts a region and returns:

- Total sales
 - Total profit
 - Total orders
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17. Technical Skills Demonstrated

This project demonstrates practical use of:

- Relational database design
 - Data transformation
 - Primary and foreign keys
 - Surrogate keys
 - Data validation
 - Multi-table joins
 - Aggregation
 - Conditional logic
 - Subqueries
 - CTEs
 - Views
 - Stored procedures
 - KPI reporting
 - Business-oriented SQL analysis
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18. Important Data Considerations

The source data contains modeling limitations that were handled explicitly during the project.

Product ID

`Product ID` was not unique in the source, so a surrogate `ProductKey` was introduced.

Location Mapping

The source location table did not contain a direct business key linking each location record to an individual order-detail record. The transformation therefore used row-order mapping as a temporary mechanism and validated the resulting foreign-key relationship.

For a production system, a stable source identifier or a clearly defined composite business key would be preferable.

19. Conclusion

The project successfully transformed raw Superstore data into a relational SQL Server database and used it to answer practical business questions.

The most important findings were:

1. **Total sales:** 501,239.75
2. **Total profit:** 39,706.21
3. **Profit margin:** 7.92%
4. **Consumer:** highest sales segment
5. **Corporate:** highest profit segment
6. **Technology:** strongest category
7. **Furniture:** negative total profit
8. **2015:** highest annual profit
9. **Houston:** highest sales city
10. **Austin:** highest-profit city
11. High discount levels were associated with substantial negative profit in the analyzed data.

Overall, the project demonstrates how SQL can be used not only to retrieve data, but to structure, validate, analyze, and interpret business information.